
HS 540: Energy: Economics, Environment and Policy

Course Code: HS 540

Course Name: Energy: Economics, Environment and Policy

L-T-P-C: 3-0-0-3

Approval: earlier approved as EN 504 Energy: Environment, Policy and Law in the 10th Senate; one-time approval as HS 540 (4 credits) in 18th BoA and 14th Senate; one-time approval as HS 540 Energy: Economics, Environment and Policy (3 credits) in 26th BoA and 18th Senate; removed as a core course of M.Tech Mechanical Engineering (Energy Systems) and kept as an elective in 48th BoA and 38th Senate

Course Outline

This course will start by reviewing fundamental concepts in engineering economics. These include concepts like estimation, supply, demand, discounting, pricing, depreciation, and simulation to capture uncertainty. Next, this course will turn towards economics of energy with a focus on concepts like payback-period, net present value, and benefit-cost analyses on energy investment projects. Finally, the last part of the course will deal with energy policy and environmental law with an emphasis to its evolution in India.

Modules

- **Module 1: Engineering Economics:** Estimation, Supply, Demand, Cost, Elasticity, Pareto Efficiency, Welfare Economics, Social Discounting, Hyperbolic Discounting, Pricing, Borrowing, Depreciation, Taxes, Market Failure, Risk Modeling, Attitudes and Utility, Multi-Attribute Decisions and Stochastic Dominance, Monte Carlo Simulation, and Cost Effectiveness Analysis/Value of Life (14)
- **Module 2: Energy Economics:** Unit cost calculation of power generation from different energy sources, Rules for investment in Energy sector, Payback period, NPV, IRR, ERR, and Benefit-cost analysis, Net Social Benefit (with Free riding concepts) and Rebound effects, Energy-GDP elasticity, Financing of energy systems, Energy – economy interaction, renewable and non-renewable sources of energy with their advantages and disadvantages on environment and climate change. (14)
- **Module 3: Energy Policy and Environmental Law:** Energy policy, Tariffs and subsidies, Taxes, National energy plan and five-year plans, Energy Models, Trend analysis, Costs of exploration and alternate energy, International energy supply, Energy Pricing, Environment interaction, Clean development mechanism, Overview of renewable energy policy in India, India's Plan for a domestic energy cap & trade scheme, and renewable energy credits. Federalism, Energy Regulation, Environmental law with a focus on Water (prevention & control of pollution) act 1974, Environmental protection act 1986, and effluent standards and ambient air quality standards. Legal Regimes Governing Primary Sources of Energy – Non – Renewable and Renewable, and Regulatory Framework concerning Electricity, Gas, Coal, and Renewable Energy. (14)